Optimizing Online Retail Sales performance using a Data-Driven Approach

A Proposal report for the BDM capstone Project

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Declaration Statement

I am working on a Project titled "**Optimizing Online Retail Sales performance using a Data-Driven Approach"**. I extend my appreciation to **UCI Machine Learning Repository** for providing the necessary resources that enabled me to conduct my project.

I hereby assert that the data presented and assessed in this project report is genuine and precise to the utmost extent of my knowledge and capabilities. The data has been gathered through primary sources and carefully analysed to assure its reliability.

Additionally, I affirm that all procedures employed for the purpose of data collection and analysis have been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures.

I am dedicated to adhering to the information of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report.

I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me. In the event that plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

I agree that all the recommendations are business-specific and limited to this project exclusively, and cannot be utilized for any other purpose with an IIT Madras tag. I understand that IIT Madras does not endorse this.

Premshuk Saha.

Signature of Candidate:

Name: Premshuk Saha

Date:12/07/2025

1. Executive Summary and Title

An established e-commerce retailer based in the United Kingdom, known for its curated selection of home decor and gift items serves to both individual shoppers and wholesale buyers on a global scale. While the business has achieved widespread reach, the company encounters challenges, it continues to grapple with operational inefficiencies that impact its growth and profitability.

The pain points include inconsistent demand, unclear insights into product-level performance, and a high rate of product returns. Unpredictable sales cycles often lead to excess inventory or stockouts, and a significant share of sales is driven by only a handful of items in the catalog. Additionally, frequent returns not only reduce profitability but also signal potential issues in product quality or customer expectations, highlighting the need to analyze return behaviors more thoroughly.

To address these issues, the proposed solution is grounded in data-analytics. The project will employ time-series forecasting models to improve demand accuracy and help maintain optimal inventory levels. A Pareto-based product analysis will be conducted to spotlight top-performing items, informing both marketing strategies and stocking decisions. Moreover, an in-depth review of return patterns will offer actionable insights into customer dissatisfaction and guide quality control efforts.

2. Organization Background

This online retail company located in the United Kingdom has been selling homeware and gifting items since the early 2000s . It offers a wide range of stylish, often exclusive products from kitchenware and décor to seasonal and personalized items that appeal to both individual shoppers and businesses. A large part of its customer base includes boutique stores and interior designers who seek unique, design-focused pieces. At the same time, it caters to everyday customers who value creative and high-quality items at competitive prices. This dual focus supports both B2C and B2B operations.

However, like many e-commerce businesses, the company faces several operational challenges. Unpredictable sales trends often lead to stockouts or overstocking. Additionally, a small portion of its products generates most of the revenue, and high return rates further impact profitability.

To address these issues, the company is shifting toward data-driven solutions. Forecasting demand using historical data, identifying top-performing products, and analyzing return patterns can provide actionable insights. With these tools, the company hopes to reduce waste, improve stock planning, and enhance customer satisfaction, ultimately supporting long-term growth.

3. Problem Statement

The business problems that could be identified from the dataset are as follows:

- **3.1 Trouble in identifying Best-Selling Products**: No reliable system to identify and prioritize high-performing products, limiting marketing and inventory optimization.
- **3.2 Difficulty in Product Demand Forecasting**: Poor demand estimates often lead to stockouts or excess inventory, affecting sales and storage efficiency.
- **3.3 Return rates are too high**: Frequent returns raise logistics costs and reduce margins, with little insight into underlying causes.

4. Background of the problem

The company is facing several key operational challenges that are limiting its growth and efficiency. These issues are affecting core areas such as inventory management, product strategy and customer satisfaction.

- One major issue is the company is unable to clearly identify which products drive the most revenue. While a small portion of the product range accounts for a large share of sales, the company lacks the data analysis needed to pinpoint and prioritize these items.

- Another challenge is its struggle to constantly predict customer demand. Sales pattern vary widely depending on the season, product type and customer group making it difficult to maintain the right level of inventory. This often leads to stockout or over-stocking which increases warehousing costs and the risk of becoming unsold products obsolete.

- Additionally, the business experiences frequent product returns due to issues like quality concerns, unmet customer expectations, or shipping problems. Without a proper system to track and analyze return trends, it's difficult to address the root causes and reduce return rates, which negatively impacts profitability and customer satisfaction.

In conclusion, finding the main reasons behind these problems is important to solve them and help the company do better.

5. Problem Solving Approach

The company is dealing with several business challenges that are slowing down its growth and reducing profitability. To overcome these problems, a structured and focused problem-solving approach is required.

i. Data Preparation:

- Gather relevant information such as past sales figures, inventory logs, and return records from the company's online retail system.
- Address data quality issues by filling the missing values, removing duplicate entries, and standardizing data formats (e.g.: dates, codes)
- Reorganize raw transaction logs into well-structured datasets that are ready for analytical modeling and interpretation.

ii. Best-Selling Product Analysis:

- Use the Pareto Principle (80/20 rule) to identify revenue-driving products.
- Classify products into high**,** medium**,** and low performers**.**
- Focus marketing and promotional efforts on top-selling items.

iii. Product Demand Forecasting using Time-Series Models:

- Apply time-series models like ARIMA, Prophet or Exponential Smoothing.
- Forecast demand at the individual product level.
- Identify seasonality and long-term sales trends.
- Use insights to plan inventory and reduce and reduce stockouts/overstock.

iv. Return Pattern Analysis:

- Analyze return data to uncover common patterns such as frequently returned products, repeat customers and high-return customers.
- Investigate reasons for returns such as quality issues, incorrect items or unmet expectations.
- Recommend improvements such as better product descriptions, enhanced quality checks, or design changes to reduce further returns.

Tools used for analysis:

The analysis of data would be conducted on Microsoft Excel using its built-in functions and tools for effective data handling. Tools such as pivot tables would be of great benefit. Additionally tools such as pie chart, bar charts, would also be used to get a good visual representation of the problems.

6. Expected Timeline

6.1 Gantt Chart:

TASK	Start	End	Duration
Data Collection	1-Jul	2-Jul	2
Data Cleansing	3-Jul	3-Jul	1
Data Analysis	4-Jul	6-Jul	3
Preparing Proposal	7-Jul	9-Jul	3
Finding problems	10-Jul	12-Jul	3
Finding solutions	13-Jul	14-Jul	2
Preparing for final report	15-Jul	18-Jul	4
Preparing for viva voice	19-Jul	20-Jul	2

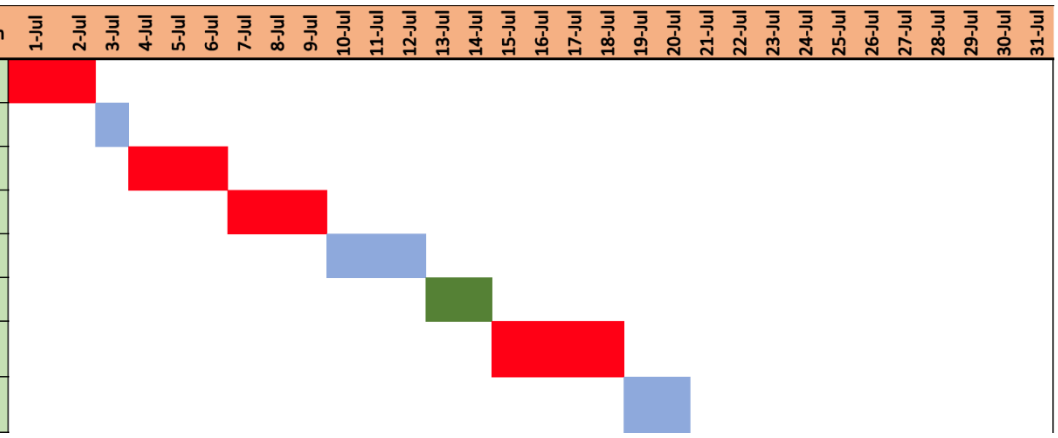

Figure 6.1. Expected timeline for completion of the Project.

6.2 Work Breakdown Structure:

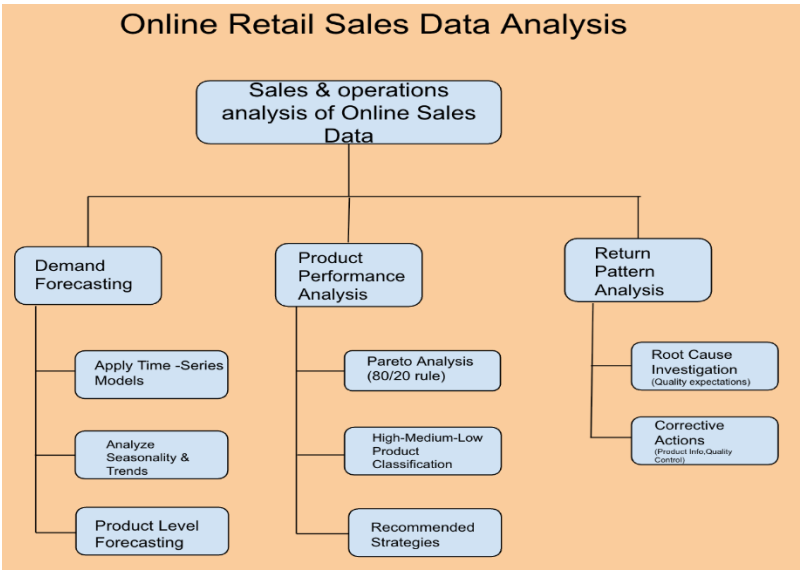

Figure 6.2 : Work Breakdown Structure

7. Expected Outcome

Implementing a data-driven approach is expected to bring significant improvements across several key operational areas of the business. The following outcomes are expected to achieve after the completion of my analysis:

1. **Clear Identification of Best-Selling Products:**
   Using techniques such as Pareto analysis, we will be able to identify which products drive the majority of revenue. This insight will allow for more focused marketing, optimized stock allocation, and data-driven decision-making around promotions and product development.
2. **More Accurate Demand Forecasting:**
   By analyzing historical sales data and seasonal trends, the company will be able to predict product demand more reliably. This will support smarter inventory planning, reducing the risk of stockouts and overstocking.
3. **Smarter Inventory Management:**
   Improved forecasting will lead to better alignment between supply and demand. This will minimize storage costs, reduce product waste, and ensure that popular items are consistently available to customers.
4. **Reduced Product Return Rates:**
   A detailed analysis of return trends will highlight which products are frequently returned and why. Addressing these issues whether related to quality, misleading descriptions, or shipping will help lower return rates, reduce logistics costs, and improve customer satisfaction.

Overall, this analytical approach will enhance operational efficiency, increase profitability, and provide a better experience for both individual customers and wholesale partners.